Question	Sum of number of protons, electrons and neutrons in 12g of 612C is:
Type	multiple choice
Option	1.8
Option	12.044×10^{23}
Option	1.084×10^{25}
Option	1.084×10^{23}
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The number of atoms present in 16 g of oxygen is
Type	multiple choice
Option	$6.02\times10^{11.5}$
Option	3.01×10^{23}
Option	$3.01\times10^{11.5}$
Option	6.02×10^{23}
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following contains maximum number of atoms?
Type	multiple choice
Option	4g of H_2
Option	16g of O_2
Option	28g of N_2
Option	18g of H_2O
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Number of oxygen atoms in 8 g of ozone is-
Type	multiple choice
Option	6.02×10^{23}
Option	$\frac{6.02\times10^{23}}{2}$
Option	$\frac{6.02\times10^{23}}{3}$
Option	$\frac{6.02\times10^{23}}{6}$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The number of molecule in 4.25 g of NH_3 is -
Type	multiple choice
Option	1.505×10^{23}
Option	3.01×10^{23}
Option	6.02×10^{23}
Option	None of these

Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Calculate the number of atoms in 11.2 litre of SO_2 gas at STP:
Type	multiple choice
Option	$\frac{N_A}{2}$
Option	$\frac{3N_A}{2}$
Option	$3N_A$
Option	N_A
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The total number of electrons present in 18 mL of water:
Type	multiple choice
Option	6.02×10^{22}
Option	6.02×10^{23}
Option	6.02×10^{24}
Option	6.02×10^{25}
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Number of neutrons present in 1.7 g of ammonia is -
Type	multiple choice
Option	N_A
Option	$\frac{N_A}{10}\times4$
Option	$\left(\frac{N_A}{10}\right)\times7$
Option	$N_A\times10\times7$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Vapour density of gas is 11.2. Volume occupied by 2.4 g of this at STP will be-
Type	multiple choice
Option	11.2 L
Option	2.24 L
Option	22.4 L
Option	2.4 L
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The number of atoms present in 0.5 g atom of nitrogen is same as the atoms in-
Type	multiple choice

Option	12g of C
Option	32g of S
Option	8g of oxygen
Option	24g of Mg
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Which contains least no. of molecules:
Type	multiple choice
Option	1 g CO_2
Option	1 g N_2
Option	1 g O_2
Option	1 g H_2
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Gram atomic weight of oxygen is
Type	multiple choice
Option	16 amu
Option	16 g
Option	32 amut
Option	32 g
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following statement is correct?
Type	multiple choice
Option	Liquids have definite volume but not the definite shape
Option	Gases have neither definite volume nor definite shape
Option	Both (a) and (b)
Option	None of the above
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The number of gram molecules of oxygen in 6.02×10^{24} CO molecules is
Type	multiple choice
Option	10 g molecules
Option	5 g molecules
Option	1 g moleucles
Option	0.5 g molecules
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	If 3.01×10^{20} molecules are removed from 98 mg of H_2SO_4, then the number of moles of H_2SO_4 left are:
Type	multiple choice

Option	0.1×10^{-3}
Option	0.5×10^{-3}
Option	1.66×10^{-3}
Option	9.95×10^{-2}
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The modern atomic weight scale is based on
Type	multiple choice
Option	C^{12}
Option	C^{16}
Option	H^{1}
Option	C^{13}
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	1 amu is equal to
Type	multiple choice
Option	$1/12$ of $C-12$
Option	$1/14$ of $O-16$
Option	1 g of H_2
Option	1.66×10^{-23} Kg
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Boron has two stable isotopes, ^{10}B (19%) and ^{11}B (81%). The atomic mass that should appear for boron in the periodic table is
Type	multiple choice
Option	10.8
Option	10.2
Option	11.2
Option	10.0
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	If an element Z exist in two isotopic form Z^{50} and Z^{52}. The average atomic mass of Z is 51.7. Calculate the abundance of each isotopic forms
Type	multiple choice
Option	Z^{50} (15%), Z^{52} (85%)
Option	Z^{50} (85%), Z^{52} (15%)
Option	Z^{50} (5%), Z^{52} (95%)
Option	Z^{50} (95%), Z^{52} (5%)
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	He has two naturally occurring isotopes He - 3 and He - 4. The average atomic mass of helium is 3.98 amu which isotope is more abundant in nature.

Type	multiple choice
Option	He -4
Option	He - 3
Option	Both are equally abundant
Option	None of these
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	An inorganic compound contains Na = 43.4%, C = 11.3% and rest is oxygen. Calculate the empirical molecular mass of the inorganic compound.
Type	multiple choice
Option	83
Option	106
Option	104
Option	90
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Calculate the empirical formula of the mineral which has $DO_2 = 46.6\%$, $A_2O = 39.5\%$ and $H_2O = 13.9\%$ (Atomic weight of A = 27, D = 28)
Type	multiple choice
Option	$2A_2O_3 \cdot DO_2 \cdot 3H_2O$
Option	$A_2O_3 \cdot DO_2 \cdot 2H_2O$
Option	$A_2O_3 \cdot 2DO_2 \cdot 2H_2O$
Option	$3A_2O_3 \cdot 2DO_2 \cdot H_2O$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A gaseous hydrocarbon is composed of 85% carbon and has molecular mass 42. Name the gaseous hydrocarbon.
Type	multiple choice
Option	Butane
Option	Acetylene
Option	Ethane
Option	Propylene
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Calculate the percentage composition of Cr in $Cr_2(SO_4)_3$.
Type	multiple choice
Option	26.5%
Option	31.5%
Option	40%
Option	22.4%
Answer	1
Solution	
Positive Marks	4

Negative Marks	1
Question	0.216 g of aluminium reacts completely with 0.852 g of chlorine to form chloride of aluminum. The simplest empirical formula mass of the compound is
Type	multiple choice
Option	98
Option	267
Option	133.5
Option	72.5
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The concentration of a solution is 6% (w/w) and the density of the solution is 2g/ml, then concentration of solution in % (w/v) is
Type	multiple choice
Option	12%
Option	18%
Option	24%
Option	30%
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	90 gram glucose is mixed with 180 gram of water. Find mole fraction of glucose
Type	multiple choice
Option	0.047
Option	0.059
Option	0.065
Option	0.072
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	What is the mole fraction of glucose in 10% w/W glucose solution?
Type	multiple choice
Option	0.01
Option	0.02
Option	0.03
Option	0.04
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A solution is obtained by mixing 200 g of 30% and 300 g of 20% solution by weight. What is the percentage of solute in the final solution?
Type	multiple choice
Option	50%
Option	28%
Option	64%
Option	24%
Answer	4
Solution	
Positive Marks	4

Negative Marks	1
Question	What will be the mole fraction of ethanol in a sample of spirit containing 85% ethanol by mass?
Type	multiple choice
Option	0.69
Option	0.82
Option	0.85
Option	0.6
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Find molarity of 28% weight by volume of KOH solution?
Type	multiple choice
Option	7 M
Option	5 M
Option	9 M
Option	4 M
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	What will be the molarity of solution containing 20 g of Barium chloride in 250 mL of solution? (Ba: 137 g τol^{-1})
Type	multiple choice
Option	0.85 M
Option	1.2 M
Option	0.38 M
Option	0.25 M
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Find molality of 3M NaOH solution if the density of solution is 1.2 g/ml.
Type	multiple choice
Option	2.7 m
Option	3.1 m
Option	2.9 m
Option	4.2 m
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	3 M of 150 ml NH_4Cl is mixed with 2 M of 100 mL of $CaCl_2$ solution and 1 M of 250 mL of $CaCO_3$ solution. The ratio of concentration of anion and cation is
Type	multiple choice
Option	3:2
Option	19:11
Option	2:3
Option	11:9
Answer	4
Solution	
Positive Marks	4

Negative Marks	1
Question	Find molarity of 20 percent weight by volume of NaOH solution if the density of solution is 1.2 g/ml.
Type	multiple choice
Option	8 M
Option	7 M
Option	6 M
Option	9 M
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	What is the percentage purity of a sample of $CaCO_3$ if 150 g of the sample gives out 44 g of CO_2 on heating?
Type	multiple choice
Option	60 %
Option	15 %
Option	50 %
Option	67 %
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Zn metal can be obtained from zinc oxide (ZnO) by reaction at high temperature with carbon monoxide. $[ZnO + CO \rightarrow Zn + CO_2]$ The carbon monoxide is obtained from carbon. $[2C + O_2 \rightarrow 2CO]$ What is the maximum amount of zinc that can be obtained from 48 gm of carbon?
Type	multiple choice
Option	8 mole
Option	4 mole
Option	2 mole
Option	3 mole
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	If the percentage yield of the given reaction is 40%, how many total moles of the gases will be produced if 8 moles of $NaNO_3$ are initially taken? $4NaNO_3 \rightarrow 2Na_2O + 2N_2 + 5O_2$
Type	multiple choice
Option	14
Option	8.4
Option	5.6
Option	7.2
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Tw successive reactions, $A \rightarrow B$ and $B \rightarrow C$, have yields of 90% and 80%, respectively. What is the overall percentage yield for conversion of $A \rightarrow C$?

Type	multiple choice
Option	170
Option	85
Option	10
Option	72
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Mass of water (H_2O) produced by the complete combustion of 630 mg glucose is:
Type	multiple choice
Option	270 mg
Option	378 mg
Option	198 mg
Option	None of these
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	2.76 gm of silver carbonate on being strongly heated yield a residue weighing:
Type	multiple choice
Option	2.16 g
Option	2.48 g
Option	2.64 g
Option	232 g
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	'X' gm of $CaCO_3$ was completely decomposed. The mass of solid residue formed is 28 gm. What is the value of 'X' (in gm)?
Type	multiple choice
Option	100
Option	50
Option	44
Option	22
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	What will be the volume of CO_2 at NTP obtained by treating the excess of $ZnCO_3$ with 0.365 gm of HCI? $ZnCO_3 + 2HCl \rightarrow ZnCl_2 + CO_2 + H_2O$
Type	multiple choice
Option	112 ml
Option	224 ml
Option	2240 ml
Option	1120 ml
Answer	1
Solution	
Positive Marks	4

Negative Marks	
Question	The one molar solution of oxalic acid is equal to
Type	multiple choice
Option	2 N
Option	1 N
Option	N/2
Option	N/3
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	What is molarity of K^+ in aqueous solution that contains 17.4 ppm of K_2SO_4 (174 g mol^{-1})?
Type	multiple choice
Option	2×10^{-2} M
Option	2×10^{-3} M
Option	4×10^{-4} M
Option	2×10^{-4} M
Answer	4
Solution	
Positive Marks	4
Negative Marks	1